

Multivariate Exploratory Data Analysis

Overview

Multivariate Exploratory Data Analysis (EDA) was conducted to examine the interaction among multiple variables within the Netflix dataset. Unlike univariate analysis, which studies individual variables, and bivariate analysis, which investigates relationships between two variables, multivariate analysis explores how several variables collectively influence patterns within the dataset.

This phase addresses the final investigation question:

What genres have increased or decreased in availability over the years?

The analysis combines **Release Year**, **Genre (Type)**, and **Number of Titles** to identify temporal trends in Netflix's content library. By tracking genre availability across multiple years, this analysis provides insights into Netflix's evolving content acquisition strategy, changing audience preferences, and long-term portfolio expansion.

To accomplish this objective, Year × Genre pivot tables, heatmaps, trend charts, stacked area charts, and growth analyses were utilized to visualize and interpret genre evolution over time.

Methodology

The following analytical workflow was adopted for the multivariate analysis:

1. Loaded the cleaned Netflix dataset.
2. Converted the **Release_Date** column into datetime format.
3. Extracted the **Release Year** from the release date.
4. Processed the **Type** column by separating comma-separated genres into individual observations.
5. Expanded the dataset using the **explode()** function.
6. Grouped the data by **Release Year** and **Genre**.
7. Constructed a Year × Genre pivot table showing the number of titles released for each genre across different years.
8. Generated an overall heatmap to visualize genre availability over time.
9. Produced a focused heatmap highlighting the top ten genres.
10. Created a multi-line trend chart to examine genre popularity across years.
11. Interpreted the resulting patterns and derived business insights.

Investigation Question 7

What genres have increased or decreased in availability over the years?

Objective

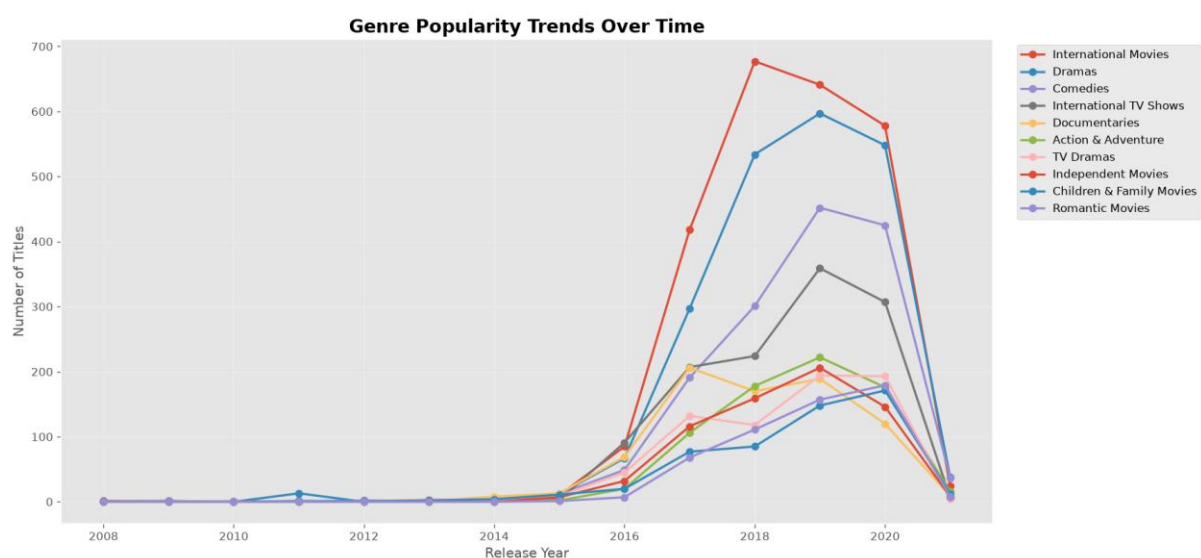
The objective of this analysis is to examine how Netflix's content portfolio has evolved over time by tracking the availability of different genres across release years. This investigation combines temporal information with genre classification to identify long-term growth trends, declining categories, and changing content acquisition priorities.

Understanding these temporal patterns provides valuable insight into Netflix's strategic investments, audience demand, and overall evolution as a global streaming platform.

Analysis Performed

The following analytical procedures were carried out:

- Extracted release years from the **Release_Date** column.
 - Split multiple genres into individual observations.
 - Expanded the dataset using the **explode()** function.
 - Created a **Release Year × Genre** contingency table.
 - Generated an overall heatmap representing genre availability across all years.
 - Created a focused heatmap for the ten most common genres.
 - Developed a multi-line trend chart to visualize changes in genre popularity over time.
 - Compared growth patterns among the major Netflix genres.
 - Interpreted temporal trends and identified long-term changes in Netflix's content strategy.
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Major Findings

The temporal analysis reveals several important trends regarding Netflix's content evolution.

- Netflix's content library expanded rapidly beginning in **2016**, with substantial growth observed across nearly every major genre between **2017** and **2020**.
- **International Movies** remained the largest genre throughout the study period, increasing from **85 titles in 2016** to a peak of **677 titles in 2018** before gradually declining to **578 titles in 2020**.
- **Dramas** consistently ranked as the second-largest genre, growing from **67 titles in 2016** to **597 titles in 2019**, demonstrating Netflix's sustained investment in narrative-driven entertainment.
- **Comedies** exhibited strong and consistent growth, increasing from **49 titles in 2016** to **452 titles in 2019**, highlighting continued audience demand for comedy-based content.
- **International TV Shows** experienced one of the fastest growth rates, rising from **91 titles in 2016** to **359 titles in 2019**, reflecting Netflix's increasing investment in serialized international productions.
- **Action & Adventure, Independent Movies, and TV Dramas** also recorded significant expansion after 2016, illustrating Netflix's strategy of simultaneously investing across multiple genres.
- **Children & Family Movies** and **Romantic Movies** continued to grow steadily through **2020**, indicating increasing demand for family-oriented and relationship-focused entertainment.
- **Documentaries** expanded rapidly between **2016** and **2017**, before stabilizing in subsequent years.
- The overall heatmap demonstrates that Netflix significantly diversified its content portfolio after **2016**, with nearly every major genre experiencing simultaneous growth.
- The noticeable decline across all genres during **2021** is most likely due to incomplete dataset coverage rather than an actual reduction in Netflix's production activities.

Business Interpretation

The multivariate analysis illustrates Netflix's transformation from a relatively modest streaming service into one of the world's largest and most diversified entertainment platforms. Beginning in **2016**, Netflix significantly increased its investment across numerous genres, reflecting an aggressive content acquisition and original production strategy designed to support rapid global expansion.

The continued dominance of **International Movies** demonstrates Netflix's commitment to international content localization, while the rapid growth of **International TV Shows** highlights the platform's increasing emphasis on serialized programming capable of improving viewer retention and long-term engagement.

Steady expansion in **Dramas, Comedies, Children & Family Movies, and Action & Adventure** further indicates that Netflix follows a balanced content acquisition strategy rather than relying on a limited number of genres. This diversified portfolio enables the platform to satisfy audiences across different age groups, cultures, and entertainment preferences.

Overall, the observed temporal trends suggest that Netflix's long-term growth has been driven by continuous diversification, strategic international investments, and data-driven content planning.

Overall Findings

The multivariate analysis demonstrates that Netflix's content portfolio evolved significantly during the study period through sustained investment across multiple genres.

The principal findings are summarized below:

- Netflix experienced rapid catalog expansion beginning in **2016**, with the highest production levels occurring between **2017** and **2020**.
- **International Movies** remained the dominant genre throughout the analysis period.
- **Dramas, Comedies, and International TV Shows** exhibited consistent long-term growth.
- Multiple genres expanded simultaneously, demonstrating Netflix's diversified content acquisition strategy.
- Family-oriented genres also experienced continuous growth, broadening Netflix's appeal across different demographic groups.
- The overall heatmap confirms that Netflix diversified its content offerings rather than concentrating on a limited number of entertainment categories.
- The decline observed during **2021** is primarily attributable to incomplete dataset coverage.

Overall, Netflix successfully expanded its content library while maintaining a balanced distribution across numerous genres, enabling the platform to strengthen its position within the global streaming industry.

Business Recommendations

Based on the multivariate analysis, the following recommendations are proposed:

1. Continue Investing in International Content

The sustained dominance of **International Movies** and the rapid growth of **International TV Shows** demonstrate strong global demand. Netflix should continue strengthening partnerships with international production companies to further expand its worldwide content portfolio.

2. Maintain Investment in High-Growth Genres

Genres such as **Dramas, Comedies, and International TV Shows** have demonstrated sustained growth throughout the study period. Continued investment in these categories is likely to maximize viewer engagement and subscriber retention.

3. Expand Emerging Genres

The consistent growth of **Children & Family Movies, Independent Movies, and Action & Adventure** suggests increasing audience demand. Further investment in these genres can broaden Netflix's demographic reach.

4. Monitor Genre Trends Continuously

Netflix should regularly monitor temporal changes in genre popularity to identify emerging audience preferences and optimize future content acquisition decisions.

5. Strengthen Data-Driven Content Planning

Temporal trend analysis should be integrated with viewer behavior analytics and recommendation systems to improve personalized content delivery and maximize audience satisfaction.

Conclusion

The Multivariate Exploratory Data Analysis successfully examined the evolution of Netflix's content portfolio by analyzing the interaction between **Release Year**, **Genre**, and **Title Availability**. The analysis identified significant growth across nearly all major genres beginning in **2016**, with the strongest expansion occurring between **2017** and **2019**.

International Movies, **Dramas**, **Comedies**, and **International TV Shows** emerged as the primary contributors to Netflix's expanding catalog, while several other genres demonstrated steady long-term growth. These findings confirm Netflix's commitment to maintaining a balanced and diversified content portfolio capable of serving a wide range of audiences across global markets.

Overall, the multivariate analysis complements the univariate and bivariate analyses by providing a temporal perspective on Netflix's content strategy. Together, these analyses present a comprehensive understanding of Netflix's content ecosystem and establish a strong analytical foundation for future predictive analytics, recommendation systems, and strategic business decision-making.